Novel Views of a Person from 125 Photos: TensoRF, 3D Gaussian Splatting and Mip-Splatting

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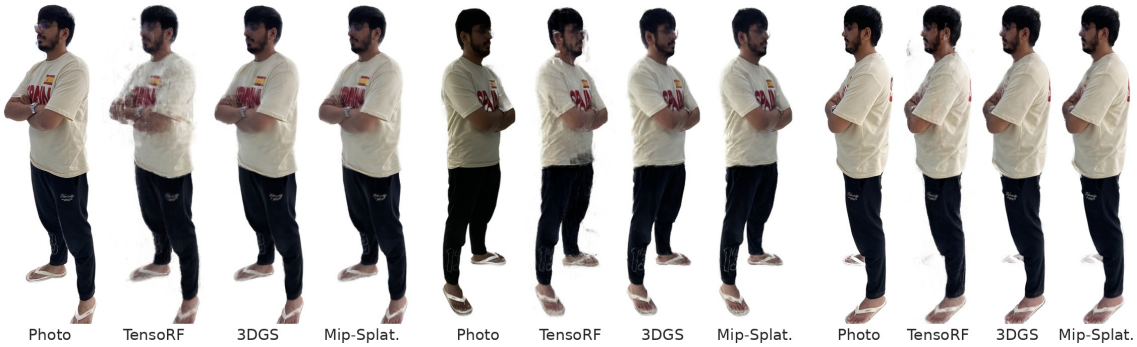


Figure 1. Three of the 16 held-out photos (not used in training) next to each method's render from the same camera.

Method	PSNR↑	SSIM↑	LPIPS↓	Train (min)	FPS↑	Primitives	MB
TensoRF [1]	26.35	0.938	0.130	42	1.0	17.4M	67
3DGS [2]	27.53	0.956	**0.064**	**7**	366	195k	46
Mip-Splatting [3]	**27.53**	**0.957**	0.064	10	**594**	217k	52

Table 1. Held-out photos (16, 750×1000): PSNR [dB], SSIM [4], LPIPS (VGG) [5], training minutes (the three ran side by side on one RTX 5000 Ada, so times include sharing), images per second, primitives (Gaussians or grid values), MB. Best in bold.

Rendered at	3DGS	Mip-Splatting
1/4 size (zoom out)	24.25 / 0.048	**28.63** / 0.028
1× (training size)	27.53 / 0.064	**27.53** / 0.064
1.6× (zoom in)	27.15 / 0.082	**27.29** / 0.081

Table 2. Zoom test: PSNR / LPIPS rendered at other sizes. Truth: photo averaged down ($\frac{1}{4}$) or the 1600 px original (1.6×).

1 Task and data

Draw a person from *new* viewpoints: a video along a preset path and an interactive viewer. Input: the 125 photos of Sub-projects 1–2 (four iPhone lenses, video frames) with the cameras Sub-project 1 recovered with VGGSfM [6], undistorted to pinhole; person masks give a white background; 1000 px. Every 8th photo (16) is held out for scoring, as in [2].

2 Methods

TensoRF [1] is a radiance field like NeRF [7]: a pixel sums density σ_i and colour c_i along its ray, $C = \sum_i T_i(1 - e^{-\sigma_i \delta_i})c_i$, $T_i = \prod_{j<i} e^{-\sigma_j \delta_j}$, but instead of one large network the 3D feature grid is factorised into planes times lines (VM decomposition). **3D Gaussian Splatting [2]** stores the scene as 3D Gaussians (position, covariance Σ, opacity, spherical-harmonics colour), started from our Sub-project 1 points and densified while training. Each is projected to a 2D Gaussian $\Sigma' = JW\Sigma W^\top J^\top$ (EWA splatting [8]), sorted by depth and alpha-blended; no ray marching, so it renders in real time. **Mip-Splatting [3]** adds a 3D smoothing filter, which limits each Gaussian's size by the training photos' sampling rate, and a 2D Mip filter that replaces 3DGS's fixed screen-space dilation, so that renders at other zoom levels do not alias.

All use the authors' code, 30k iterations, white background. Extra: 3DGS on the full photos with the room (19.71 dB).

3 Results

Table 1, Fig. 1: the Gaussian methods are the most faithful (27.53 and 27.53 dB) and draw at 366 and 594 images per second; TensoRF reaches 26.35 dB and, marching rays, draws far slower. At the training size 3DGS and Mip-Splatting are on par. Table 2 tests the claim of [3]: at a quarter of the size (camera moving away) 3DGS falls to 24.25 dB, its Gaussians thinner than a pixel, while Mip-Splatting keeps 28.63 dB; zoomed in (1.6×) both stay close.

4 Interactive viewer in the starter code

The course's Open3D [9] GUI of Sub-project 1 gains a novel-view panel, like Nerfstudio [10]: the 3D view's camera is converted to a pinhole camera, the selected model is rendered from it on a worker thread and shown behind the scene, so dragging, panning and zooming move the novel view (preview while moving, full size when still). It shows the held-out photo for comparison, exports MP4s along a preset orbit or user keyframes (Catmull-Rom positions, slerp rotations), and rasterises the Sub-project 2 meshes (Screened Poisson [11]) in the same view. To run on a laptop without CUDA we re-implemented the 3DGS/Mip-Splatting rasteriser and TensoRF's ray marching in plain PyTorch (CUDA, Apple MPS or CPU); they match the official renders at 106 and 104 dB PSNR (identical up to rounding).

References

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